Shuaiting Li

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Education

University College London *Sept 2024 – Present*
BSc Computer Science

- **First Class (Overall 84%)**
- Discrete Mathematics: **100%**

University College London *Sept 2023 – June 2024*
Undergraduate Preparatory Certificate

- Overall score: 89% (Top 1% of cohort)
- Elected Academic Representative for Mathematics

Research Interests

Compiler infrastructure and optimization · HW/SW co-design · Program analysis · MLIR/LLVM ecosystem

Experience

Industry Exchange Network Project — Lead Backend Developer *London, UK*
NTT DATA (via UCL IXN, COMP0016) *Oct 2025 – Mar 2026*

- Led backend development on a 4-person student team delivering **Cresco**, a RAG-powered AI assistant for UK small-scale farmers, sponsored by **NTT DATA**
- Designed and built the backend: **FastAPI** service orchestrating **LangGraph** agents, **ChromaDB** vector store, and **PostgreSQL**; deployed to **Azure** via **Docker**
- Delivered all 27 client requirements; led weekly client-facing demos and requirement reviews

Software Engineering Intern — Central Engineering, Systems *Cambridge, UK*
Arm *Jun 2025 – Sep 2025*

- Co-built a Python-based configuration validator and end-to-end test framework for the next-generation Generic Interrupt Controller (GIC) in Arm System IP (confidential, unreleased), working with a tech lead and another intern
- Reduced per-test runtime by approximately **60%** (10s to 4s) through profiling-driven optimizations and streamlined orchestration
- Authored **SystemVerilog** testbenches to generate stimuli and integrated validator flows with **pytest**
- Achieved **96%** code coverage using **coverage.py**; added regression runs and coverage tracking
- Participated in rigorous code reviews via **Gerrit** and contributed internal documentation

Assistant Engineer Intern *Kunshan, China*
GETAC *Jul 2024 – Aug 2024*

- Collaborated with senior engineers to develop, test, and deploy generative AI models, optimizing administrative workflows for 30+ non-technical administrative staff
- Built a **WinForms** UI in **C#**; implemented a **Python** backend integrating the **OpenAI API** and other services
- Gathered requirements from non-technical stakeholders and translated them into developer-ready documentation

Technologies

Proficient: C, C++, Python, SystemVerilog

Familiar: Java, Haskell, JavaScript, SQL

Tools: Git, Docker, LLVM/MLIR, Azure, PostgreSQL, FastAPI, LangGraph, React, pytest, Gerrit

Projects

xDSL — MLIR Compiler IR Framework *Open-source contribution*

- Contributed to xDSL, an open-source Python-based MLIR dialect framework
- Landed **2 merged PRs**: RISC-V XOR canonicalization pass (`(x^a)^a -> x`) and IR test coverage improvements with clearer error messages
- Tools used: Python, MLIR

C Compiler *Work in Progress*

- Implemented a C compiler using Bison and Flex, generating Koopa IR as an intermediate representation
- Built a backend to translate Koopa IR into RISC-V assembly, covering parsing, semantic analysis, and code generation
- Tools used: C++, Bison, Flex, Koopa IR, RISC-V

LLVM Passes *Personal Project*

- Implemented custom LLVM optimization passes built on top of LLVM's analysis framework, including loop analysis passes
- Completed the *Kaleidoscope* frontend tutorial end-to-end (lexer, parser, IR generation, JIT execution) to deepen understanding of LLVM internals
- Tools used: C++, LLVM, CMake